

Heuristic Usability Evaluation

OOPP Group 1

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1 INTRODUCTION

This report documents an evaluation of the user interface design of our OOPP team's project, with the aim of identifying and resolving any usability issues. Following the methods outlined in [3], we evaluated a prototype of our task management application, Talio. In this report, we will provide an analysis of the application's evaluated state, its problems, and the redesigned UI. To familiarize the reader with our prototype, we will now illustrate the different windows that a user may have encountered during testing.

The application starts-up with the screen in figure 1, providing two options: clicking the "User log in" button leads to the screen in figure 2, while clicking the "Admin log in" button leads to the one in figure 3. Both of them allow connecting to a desired server and returning to the previous screen, but the admin is additionally required a server password.

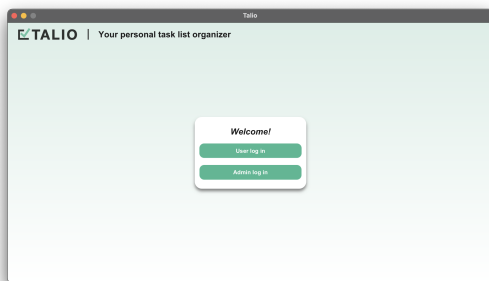


Figure 1: Start-up screen

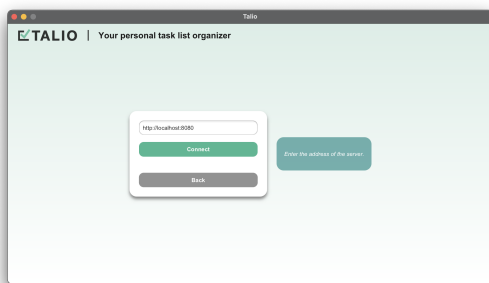


Figure 2: User log in

After connecting to the server, the workspace corresponding to the chosen role appears. An example user workspace is shown in figure 4. Both user and admin variants look and behave similarly,

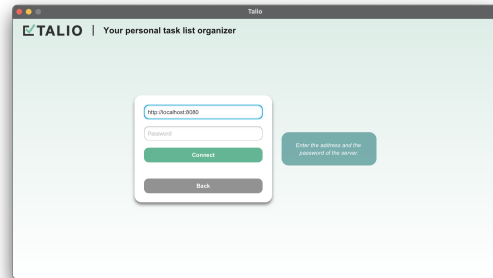


Figure 3: Admin log in

with two differences: 1) upon initialization, the user is shown no boards in the overview (since they haven't yet created any), while the admin is provided with all existing server boards; and 2) in the left-side board list, the user sees "door" icons to leave boards, while the admin sees "trash bin" icons to delete server boards.

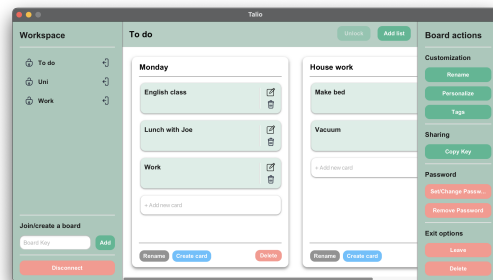


Figure 4: Workspace

The workspace presents a number of features, via the buttons in the right-side of the screen: adding a new list, renaming the board, personalizing it, creating tags which can be added to cards, copying the key of the board, setting/ changing/ removing the password of the board, and finally leaving or deleting the board.

Clicking the "Add list" button adds a new list titled "New list" to the board directly. Clicking the "Rename" button on any list makes the title editable, allowing name changes by pressing enter to confirm (figure 5).

Clicking the "Delete" button of any list opens a pop-up similar to the first one in figure 6, to confirm the deletion. The white "Add new card" button below the final card in any list can be used to quickly add a simple task by just providing a title. Alternatively, the blue "Create card" button opens a pop-up for adding a more complex card (7).

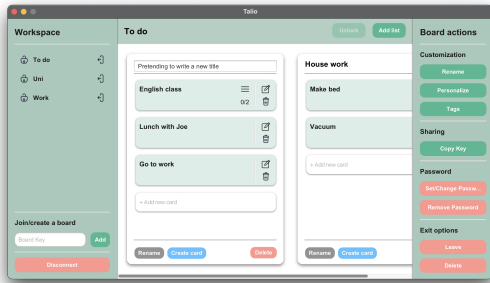


Figure 5: Renaming lists

Each card may also display indications that it has, for instance, a description or nested sub-tasks directly in the overview. The "pencil" and "trash bin" icons in the right-side of any card can be used to edit the card (which opens a window similar to the one in figure 7), or delete it, respectively.

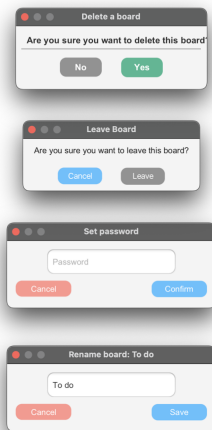


Figure 6: Pop-ups with different purposes

Double-clicking a card pops-up a window similar to the one in figure 8, allowing you to view its details.

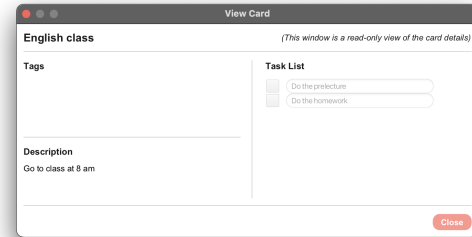


Figure 8: View a card

The "Tags" button pops-up the "Tag Overview" window, seen at the top of figure 9, which shows a list of all existing board tags, allowing you to edit, delete or add tags. The "Add a new tag" button redirects to the "Add Tag" window, seen at the bottom of the same figure.

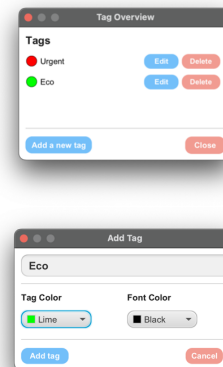


Figure 9: Tag windows

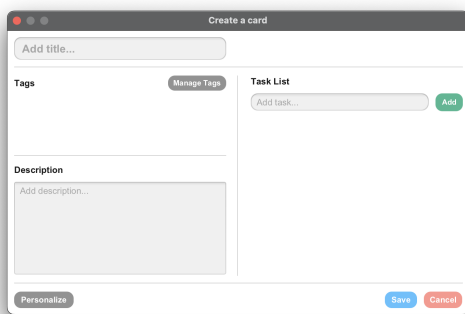


Figure 7: Create a card

The "Personalize" button opens a customization pop-up (figure 10), which allows you to change the colors of boards and lists, reset them, and also add color presets for cards.

2 METHODS

In this section, we will describe the methodology used when executing the evaluation.

2.1 Experts

We recruited 4 experts from OOPP Group 29. As Computer Science and Engineering students who have used other scheduling apps extensively, they are experienced with software such as Talio as both users and professionals, therefore being perfect choices for our evaluation.

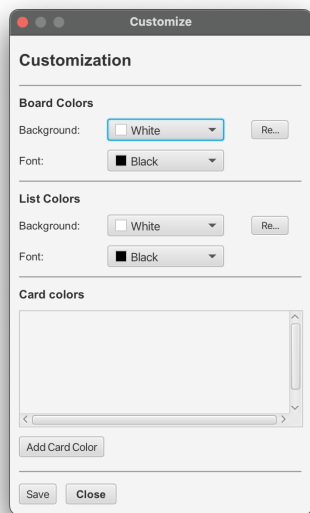


Figure 10: Customization window

2.2 Procedure

We held the procedure in the Drebbelweg building, an open environment at TU Delft, thus proper for our evaluation, which consisted of two parts.

2.2.1 First stage. During this phase of the evaluation, every expert was guided by a member of our team, who provided them with a Google Form with 8 questions focused on basic capabilities of our prototype, each of them accepting an answer from a scale of 1 to 5 about how difficult each task is (1 was marked in the form as "clear and intuitive", while 5 as "difficult and annoying"). The questions were read to each evaluator by their assistant, to make sure that all instructions are clear:

- (1) Connect to the default server as a user.
- (2) Create 3 boards.
- (3) Rename the last board you created.
- (4) Add 3 lists to this board.
- (5) Rename the last list you created.
- (6) Add 3 cards to this list.
- (7) Drag and drop the cards you created between lists on the board. How difficult is this task?
- (8) How consistent, descriptive, and minimalist would you describe the overall interface to be?

Each evaluator had to answer the questions individually to minimize biased results, and in the case of an answer of at least 3 (i.e., poorly implemented feature), the evaluators were also prompted to motivate their choice.

2.2.2 Second stage. After going through the typical usage scenario presented in stage 1, the experts were now asked to test the app freely, for a period of 1 hour, and to describe the problems they found in another Google Form, containing clear instructions and rules devised for replicability purposes. They were required to

report the found problems in the following format, which we took from [1]:

- (1) Problem description: a brief description of the problem;
- (2) Likely/actual difficulties: the anticipated difficulties that the user will encounter as a consequence of this problem;
- (3) Specific contexts: the specific context in which the problem may occur;
- (4) Assumed causes: description of the cause(s) of the problem.

Lastly, we required the evaluators to follow the 10 heuristics described in [2] when writing the "Heuristic" of each problem. For this, we made sure that the experts were familiar with the 10 heuristics beforehand, and we included them in the form so that they could have a quick reference at them. Due to word limitations, we will only mention the titles of the heuristics here, but they are all detailed in [2]:

- (1) Visibility of system status;
- (2) Match between system and the real world;
- (3) User control and freedom;
- (4) Consistency and standards;
- (5) Error prevention;
- (6) Recognition rather than recall;
- (7) Flexibility and efficiency of use;
- (8) Aesthetic and minimalist design;
- (9) Help users recognize, diagnose, and recover from errors;
- (10) Help and documentation.

In situations where the experts struggled with the features, we gave them hints of how to achieve what they want, and we also noted down any comments from them, without responding.

2.3 Measures (Data collection)

Raw results were collected via Google Forms automatically. Additionally, our team recorded any other comments/remarks from the experts on paper.

3 RESULTS

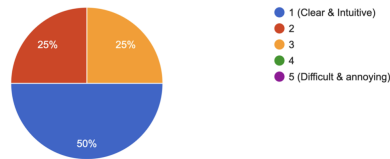
In this section, we will list and analyze the identified problems, based on the two evaluation stages, while also prioritizing them accordingly.

3.1 Stage 1 problems

We processed the results of this stage by taking into consideration the form questions with responses of difficulty at least 3. These are shown in figures 11 and 12, which helped us extract the following problems:

- **Problem 1:** Adding lists with the predefined "New list" title is inconvenient as it requires an extra step to rename it to your desired title (heuristic: "Flexibility and efficiency of use");
- **Problem 2:** It is not explicit that users have to press Enter to save the new title when renaming a list, and moving the cursor doesn't allow pressing Enter to save anymore (heuristic: "Recognition rather than recall").

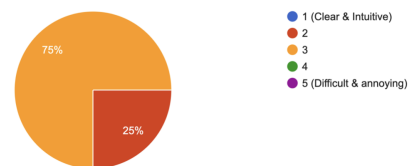
Add 3 lists to this board. How difficult is this task?
4 responses



Motivation from the evaluator that gave 3: I believe that when I add a list I would like to chose its name so that I do not have to rename it afterwards. Adding itself is not difficult, but this is my only concern.

Figure 11: Results of question (4)

Rename the last list you created. How difficult is this task?
4 responses



Motivations from the evaluators that gave 3:

- If you write a new name and you move your cursor, you can't press enter to save.
- I do not have a button so save the name of the list, since I didn't know that I have to click enter.
- It is quite unclear how to actually finish renaming the list, and I also believe there should be a cancel functionality. In a way, I would assume a user would like to cancel this action but also to confirm it. Now it is not clear that entering saves.

Figure 12: Results of question (5)

3.2 Stage 2 problems

Here the collected problems were processed as follows: we took those functionality-related out of consideration as they were out of scope for our evaluation, removed duplicates and split mistakenly grouped problems.

Problem 3

Identified Heuristic: Consistency and standards

- (1) Problem description: The board key is the same as the title;
- (2) Likely/actual difficulties: The user may be confused as to whether the title is also the key of the board or they are two distinct entities;
- (3) Specific contexts: If you rename the board to something completely unrelated, when sharing the key, it might cause confusion (a board having "Company" as the invite key can be renamed to "Personal");
- (4) Assumed causes: Not enough consideration for people who are unfamiliar with scheduling apps.

Problem 4

Identified Heuristic: Match between system and the real world

- (1) Problem description: Users might be confused by the "Task List" section inside of a task;

- (2) Likely/actual difficulties: The card is already added to a list, so the user can get confused as to which Task List the label is referring to;
- (3) Specific contexts: Whenever the user views or edits a card;
- (4) Assumed causes: Oversight during design.

Problem 5

Identified Heuristic: Help and documentation

- (1) Problem description: There is no help screen to help me understand the app;
- (2) Likely/actual difficulties: I may have to go through multiple screens if I want to achieve a frequent task, like adding a tag to a card;
- (3) Specific contexts: When I want to add a tag, I first have to open the card editing pop-up by mouse, then the manage tags pop-up, and then I can pick a tag. While this is fine by mouse, it could be quicker with a shortcut.
- (4) Assumed causes: Not providing a help screen of frequently-used shortcuts.

Problem 6

Identified Heuristic: Consistency and standards

- (1) Problem description: Many pop-up screens have different designs.
- (2) Likely/actual difficulties: It can be difficult to adapt to the way different screens look at first glance.
- (3) Specific contexts: The customization window lacks the same colors that other windows have, and when deleting or leaving a board you get different pop-up styles.
- (4) Assumed causes: Not taking consistency into consideration.

3.3 Other problems

Based on the comments of the evaluators and the tendencies we noticed in them while using the UI, we further found these problems:

- **Problem 7:** Unfamiliar users may spend time identifying buttons in the right side of the workspace (heuristic: "Recognition rather than recall");
- **Problem 8:** The positioning of buttons on the right side of pop-ups is inflexible, as we noticed that most of the evaluators had a tendency of clicking to the left (heuristic: "Flexibility and efficiency of use").

3.4 Prioritization

After a team discussion, we established the severity of a problem based on 2 criteria: the frequency and the impact. Each member individually assigned a weight from 1 (which meant "very low") to 5 (which meant "critical") to these two criteria for every problem. Then we computed the mean frequency and impact of every problem and rounded these to the nearest integer in the range from 1 to 5, obtaining the final weights.

The prioritization can be seen in figure 13. We addressed the problems with the highest frequency first, since this meant they were found in multiple parts of our app so they had to be resolved

quickly. In the case of equal frequencies, we addressed the problems based on their impact. Else, we treated the problems equally.

		Frequency				
		1	2	3	4	5
Impact	1					
	2		P5	P1, P7	P8	
	3		P3, P4	P2		P6
	4					
	5					

Figure 13: Severity matrix of the found problems

4 CONCLUSIONS & IMPROVEMENTS

4.1 Conclusions

We will now derive a conclusion based on our results.

Problem 1: This is indeed the case. When creating a new list the default name is "New list", which forces users to rename them if they wish for another name, which can get tiresome quickly.

Problem 2: This is also a good catch. In our design, when pressing the rename button on a list a text field becomes visible. The user can then type a new name and press enter to save. It is, however, unclear that the user must press enter for the changes to take effect. Moreover, there exists no cancel button, meaning if the user changes their mind they can not revert their changes.

Problem 3: This is a good observation we missed: the user might want to input only the board name and not have to worry about the key, but right now the board name is the key by default. This not only makes it harder for new users to understand the difference between board keys and board names but it is also harder to connect to boards via keys if the name is changed and the user believes that the key has changed as well.

Problem 4: This is another mistake we made: the user will most likely get confused as to what the label "Task List" actually refers to, when it is referring to sub-tasks. This is due to internal vocabulary: during development we referred to "Tasks" as "Cards" and "Sub-tasks" simply as "Tasks". It is easy to see how the confusion was created, especially when most people use the terms "Tasks" and "Cards" interchangeably during the course.

Problem 5: This problem is simple: there does not exist a "Help" screen for the user to access. This can be very troublesome for inexperienced users, which might feel overwhelmed at first.

Problem 6: Another mistake that slipped through the cracks: pop-ups have slightly different designs that make it harder for the user to adapt to the app.

Overall, our user interface proved to lack a sense of consistency, flexibility, and also directions related to how exactly the application works for new users. We will now discuss our final design, explaining why we applied certain changes compared to our initial prototype.

4.2 Improvements

As seen in figure 14, we have redesigned the workspace to accommodate a new text field for the board key and a new button that allows the user to join an already existing board. Now the user can not only clearly see and understand the difference between a board key and a board name, they can create a new board by providing only the name, while the key is generated automatically. Adding the "Join" button makes it clear that the "Add" button's functionality is creating new boards, thus removing the confusion expressed in Problem 3. In this way we have improved our design as required by heuristic 4 (Consistency and standards) and heuristic 2 (Match between system and real world).

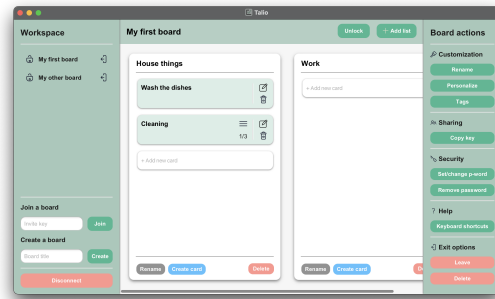


Figure 14: New main screen

As pertains to problems 1 and 2, as can be seen in figure 15, we have added two new buttons when the user wants to rename a list: a "Save" button that commits the changes and a "Cancel" button that keeps the current name. In this way we address the concerns raised by the second problem and we improve our user control and freedom (heuristic 3).

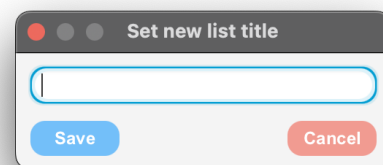


Figure 15: Improved List management

Regarding the complaints raised in Problem 4, we have redesigned the "View Card" and "Edit Card" screen (figure 16): now, the label reads "Sub-task List" making it clear that this list refers to minor steps that have to be completed in order to finish the larger problem for which the card was created. This change is in accordance with heuristic 10, help and documentation.

In order to address problem 5 we implemented a new Help window that presents the user with information regarding the

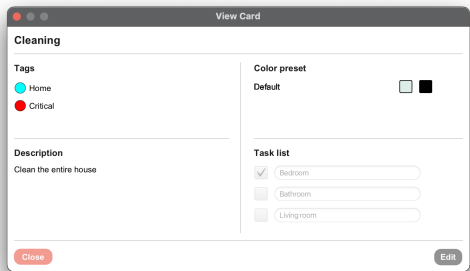


Figure 16: Updated "View Card" screen

app (figure 17), such as the functionality of certain buttons and shortcuts. By doing this we improve yet again the level of help and documentation in our app (heuristic 10).

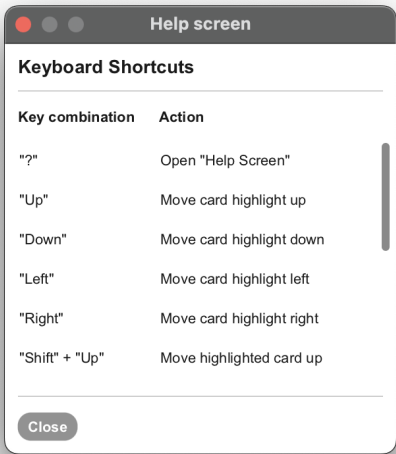


Figure 17: Help Window

Finally, in order to address problem 6 we have reviewed the design of the entire app, making adjustments were needed with a heavy focus on consistency.

Taking everything into account, the evaluation allowed us to do a wide range of improvements to the design, in conformity with our heuristics. We thank the members of OOPP Team 29 for assessing our prototype and giving us constructive feedback.

REFERENCES

[1] Gilbert Cockton, Alan Woolrych, Lynne Hall, and Mark Hindmarch. 2004. *Changing Analysts' Tunes: The Surprising Impact of a New Instrument for Usability Inspection Method Assessment*. 145–161. https://doi.org/10.1007/978-1-4471-3754-2_9

[2] Jakob Nielsen. 1994. *Heuristic Evaluation*. John Wiley & Sons, 25–62.

[3] Jakob Nielsen. 1994. How to Conduct a Heuristic Evaluation. <https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation>. Accessed: 2023-24-03.